

BEE 208 — Question 13: Earth Resistance Test

Control Statement: while loop | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variable resistance (double)

Step 3: Prompt user to enter first resistance reading

Step 4: Read resistance

Step 5: While resistance $\leq$ 5, display 'Earth resistance is acceptable'

Step 6: Prompt and read next resistance reading

Step 7: When resistance $>$ 5, exit loop

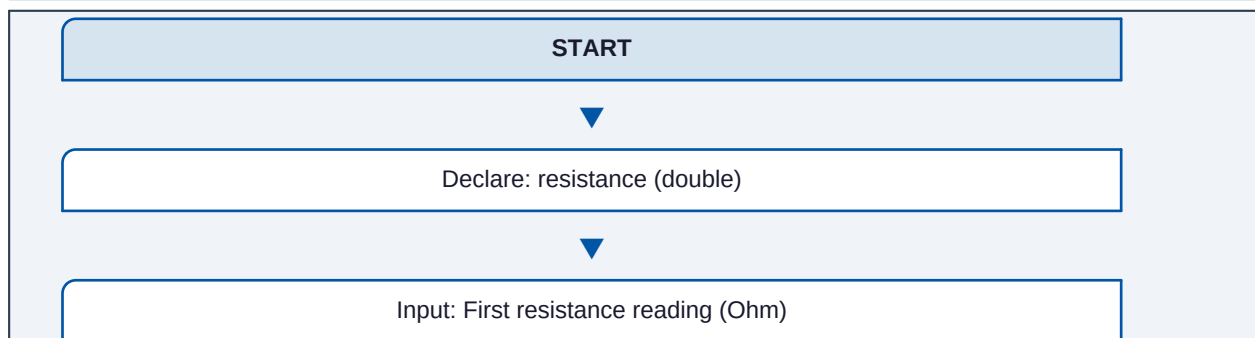
Step 8: Display 'Earth resistance too high. Improve earthing system.'

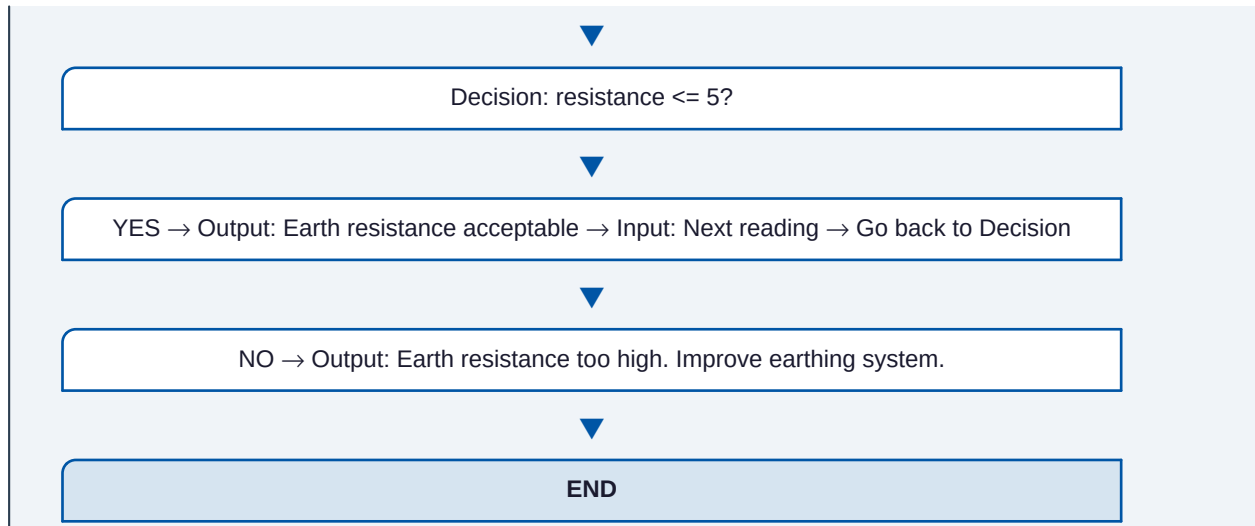
Step 9: End

2. PSEUDOCODE

```
BEGIN  
  
DECLARE resistance AS DOUBLE  
  
OUTPUT 'Enter first resistance reading (Ohm):'  
  
INPUT resistance  
  
WHILE resistance  $\leq$  5 DO  
    OUTPUT 'Earth resistance acceptable:', resistance, 'Ohm'  
    OUTPUT 'Enter next resistance reading (Ohm):'  
    INPUT resistance  
END WHILE  
  
OUTPUT 'Earth resistance too high. Improve earthing system.'  
  
END
```

3. FLOWCHART (Structured Diagram)





4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	3, 4, 6	Expected Output	OK×2, then: Earth resistance too high. Improve earthing system.
Input	8 (first entry)	Expected Output	Immediately: Earth resistance too high. Improve earthing system.